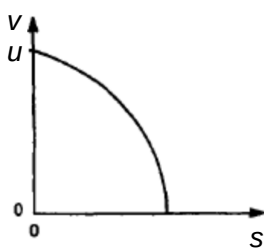
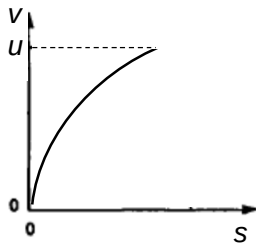


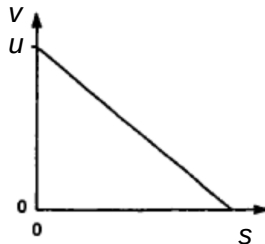
- 4 A falling stone strikes some soft ground at speed u and suffers a constant deceleration until it stops. Which one of the following graphs best represents the variation of the stone's speed, v , with distance, s , measured downwards, from the surface of the ground?



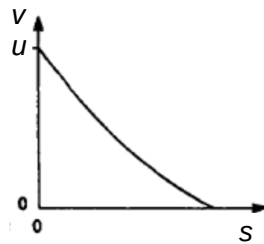
A



B



C



D

$v^2 = u^2 + 2 a s$ where a is negative.

$$\frac{dv}{ds} = \frac{dv}{dt} \times \frac{dt}{ds} = \frac{a}{v}$$

As v decrease, gradient of v - s graph becomes increasingly negative.

- 5 A trolley of mass 5.0 kg travelling at a speed of 6.0 m s^{-1} collides head-on and locks together